
The Demand for Environmental Quality: An Application of Hedonic Pricing in Golf

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Frank F. Limehouse¹, Peter C. Melvin², and Robert E. McCormick³

Abstract

The analysis uses an exhaustive golf course database that contains over 100 golf course variables on more than 15,000 golf courses in the United States combined with data from the Audubon International Cooperative Sanctuary Program to examine the market setting of environmental certification on golf courses. Using the Rosen (1974) two-stage estimation technique, quality-adjusted structural demand and supply equations for golf are estimated. A standard hedonic pricing model shows a substantial price premium for environmentally certified Audubon International golf courses. Additional results suggest that the increase in marginal benefits of certification is approximately equal to the increase in marginal costs.

Keywords

environmental economics, golf, hedonic pricing, environmental certification

¹ Chicago Census Research Data Center and DePaul University

² Ernst and Young LLP

³ Clemson University

Corresponding Author:

Frank F. Limehouse, Chicago Census Research Data Center, 230 South LaSalle Street, Chicago, IL 60604
Email: flimehouse@frbchi.org

Introduction

This article is a contemporary, detailed look at golf courses in the United States. At present, there are more than 17,000 golf courses in the United States with approximately 1,400 new golf courses opening in the past 5 years. A recent National Golf Foundation report estimates there are approximately 27 million golfers in the United States accounting for approximately 10% of the total population. In 2002, the average 18 hole green fee (including cart) was approximately US\$59.00.

We investigate willingness to pay for improved environmental quality. We do this in the context of golf using quality-adjusted empirical demand and supply estimates. This approach allows us to assess the market demand for measurable environmental quality certification. The first step in this process estimates a golf course hedonic pricing model using the approach suggested by Rosen (1974). Normally, hedonic estimation only focuses on reduced form estimates of quality characteristics, and this simplification is most often attributable to the absence of quantity data. We overcome this problem with price, quality, and quantity data on golf courses. Following Rosen, we first assign a dollar value to each observable golf course characteristic using a single product quality equation. We then use the quality-adjusted prices from the hedonic model to simultaneously estimate demand and supply equations for golf.

Our approach combines two useful, independent strands of literature: the hedonic estimation technique with the demand for environmental quality. To some, golf courses are substantial environmental liabilities and are often criticized because they degrade water quality with runoff, use excessive amounts of water, use large amounts of chemical applications for fertilizer and pest control, and destroy natural wildlife and their habitat.¹ The Audubon certification system addresses these issues by encouraging certified golf courses to enhance their practices in ways that create environmental assets and value. Courses may freely choose to become certified by the private, nonprofit organization, Audubon International, and insofar as we know there is no mandate that requires courses to seek certification.

We note that environmental certification may not be valued directly by golfers *per se*, but it may be also demanded or urged by third parties and thrust upon golfers with limited options. Whether environmental alteration of golf course design and operation is demanded by golfers because they like a more natural environment or value a more environmentally friendly course is not the central issue addressed in this analysis. The working idea is that golfers or external third parties want golf courses to be more environmentally conscious. The focus of our analysis is on the impact of the demand and supply of courses deemed environmentally altered to meet certain habitat and pollution standards. Golf courses supply the altered golf environment and a third party, Audubon International, sets the standards and meters course compliance.

In the next section, we provide a brief background into environmental issues on golf courses and potential solutions. In the third section, we discuss the hedonic

literature and how it relates to our model. Our data discussion and empirical results are in the fourth section, followed by a conclusion.

Golf Courses and Environmental Issues

Potential environmental problems exist in golf course operations in three major areas: chemical contamination, water conservation, and wildlife habitat. Since the early 1980s, chemical use on golf courses has been a major environmental concern.² The main issue with chemical use on golf courses is damage to adjacent water sources from chemical runoff. Golf courses are also criticized for irrigating in areas where water resources are particularly scarce. Golfers demand good-looking turf and when drought hits, the problem becomes twofold—water becomes more expensive and the turf requires more water. The toll on wildlife is another area where course owners feel pressure from environmentalists. Pesticides often kill insects that other species use for food. Environmentalists also question the cookie cutter approach to golf course design and its impacts on wildlife migration and habitat. That is, golf holes are often designed where they cut through essential areas of wildlife habitat, thus not completely eliminating the habitat but placing constraints on its use.

Market opportunities may exist when one party's action affects another party's welfare and the relevant costs or benefits are not reflective in the marketplace. In the case of golf courses, these opportunities exist if golf course construction results in the elimination or reduction of wildlife habitat in areas where golfers or other people place a high value on the wildlife. This value may come in the form of seeing or hearing wildlife or simply knowing that wildlife has a healthy habitat. In either case, some environmentalists argue that golf courses, in effect, use the environment in ways that do not compensate external parties for the loss of wildlife habitat or other environmental amenities. This problem, as in the tragedy of the commons, has its roots in poorly defined property rights. This creates a situation where contracting can create value for golf course owners, golfers, and wildlife lovers. If golf course operators can find a way to charge golfers for enhanced wildlife habitat and golfers would prefer to play courses amenable to wildlife, then altering courses to make them environmentally friendly will increase play and allow for higher fees. As environmental demand increases and as population and crowding grow, a new market potential has emerged to use golf courses as wildlife refuges and green space to complement land development.

Audubon International attempts to convince golf courses that being certified and enhancing environmental stewardship creates value for the course. Certification by a third party helps ensure that environmental actions are appropriate and consistent across golf courses. That is, certification provides benefits to golf courses by publicly demonstrating and communicating environmental stewardship activities. Courses may then alert golfers, environmentalists, or other parties to their environmental activities.³

The value of golf course certification may come in several forms. First, golfers may simply want to play on a course that is environmentally friendly. For example, they may enjoy having naturalized habitat areas on the course that were once maintained turfgrass (as specified in the Wildlife Habitat and Management component of Audubon certification). In this case, the golf course must identify the demand for the naturalized area and also consider the costs of conversion. The cost of turfgrass conversion simply come in the form of the land's next best alternative—maintained turf. The course must consider the forgone value of having a lush fairway that golfers also value. For courses such as Augusta National Golf Club whose members seem to place a high value on having almost perfectly manicured fairways and greens, certification is probably too costly. We are not surprised that there are differing demands for golf course landscaping. In addition to the demand by golfers, home owners may place a high value for land adjacent to golf courses with certain environmental attributes. Thus, golf course owners may capture benefit from certification either through higher green fees or land rents obtained by selling real estate adjacent to environmentally friendly golf courses.⁴

The value of certification may also come in the form of avoiding criticism from environmentalists.⁵ Some courses may also be required to gain certification if they are to receive construction permits. Although we have not observed such regulatory requirements for golf course construction projects in the United States, several counties and townships abroad are considering requiring new courses to be built under the Audubon International Signature Golf program (J. Zeh, personal communication, 2003). Thus, the owners of some courses may feel pressure from regulatory agencies to become certified or to avoid future regulation. Whether demand is driven by golfers who desire to play an environmentally friendly round or by golf courses who want to demonstrate their stewardship activities to environmentalists, Audubon has recognized this demand and has acted to capture its value.

The Audubon International Cooperative Sanctuary Program gives course managers the information and guidance to improve environmental quality (Audubon International, 2000 and 2002).⁶ A golf course may gain certification in one of six environmental areas: environmental planning, wildlife habitat, chemical use reduction, water conservation, water quality, and outreach and education. A course is required to first complete the environmental planning component to become a certified registered member. Once a course completes projects in all of these areas, it receives recognition as a Certified Audubon Cooperative Sanctuary as well as a Certified Audubon Cooperative art display, logo, and press release. Every two years, certified courses must submit recertification materials to ensure certification standards are maintained. In addition, member courses must submit one case study describing a particular environmental project on the course, such as the naturalization of maintained turfgrass. This case study must be submitted before the last component of certification is approved by Audubon International. A brief description and the requirements of each area of the program are reported in the Appendix.

In addition to the Cooperative Sanctuary Program, Audubon International offers the Signature Cooperative Sanctuary Program, which provides environmental stewardship assistance to landowners and developers from the planning stages of project design. The program requires members to develop a Natural Resource Management Plan under the review and approval of Audubon. Components of the plan include site assessment, habitat enhancement, waste management, energy efficiency, water conservation and quality, and integrated pest management. There are three Audubon Signature membership designations: Bronze, Silver, and Gold. In addition to submitting an acceptable Natural Resource Management Plan, a Bronze member golf course must host site visits during construction, pass periodic site audits, and submit periodic reports to Audubon International. A Silver Signature member golf course is held to the same requirements but the Natural Resource Management Plan is completed by environmental planners at Audubon. The Gold Membership golf course has all aspects of the property prepared by the Natural Resource Management Plan, including architecture, infrastructure, landscaping, and community education.

Our analysis attempts to compare costs and benefits of participating in the Audubon certification program across the various levels of course certification. For example, the registration fee of the Audubon Signature Status Program is US\$9,500, compared to the US\$150 annual membership fee for the Cooperative Sanctuary Program. Certain certification categories are more costly than others and costs of certification will also vary across the characteristics of different courses. A golf course already practicing the environmental principles of the program will find it less costly than a course that must invest marginal time and money to become familiar with program requirements. For example, adopting the water conservation component of certification for a course with a technologically advanced and computerized irrigation system should be relatively inexpensive.⁷

There has been other work in the area of environmental certification. Nimon and Beghin (1999), for example, estimated hedonic price functions to identify price premiums of environmentally certified fibers and dyes in the apparel industry. Nimon (2002) examined financial risks to farmers of producing organic cotton. Teisl, Roe, and Levy (1999) questioned the notion that businesses receive value-added rents from emphasizing ecocertification or ecoseals. O'Brien and Teisl (2002) found significant price premiums for environmental attributes of certified forest products. Bockstael and Strand (1997) examined the efficiency of dolphin-safe tuna labels as an instrument for environmental preferences.

Hedonic Literature

Since Working (1927), hedonic pricing models have been used in myriad ways. Waugh, in Brendt (1991), attempted to determine the implicit prices of asparagus based on characteristic bundles (i.e., width, number of stalks, and color). Other

prominent scholars have contributed to the theory and understanding of hedonic pricing.⁸

Rosen's (1974) seminal paper most aptly applies to our analysis of golf courses and environmental certification.⁹ Rosen's model brings together producers and consumers in a spatial context. The producers develop goods that contain final consumer characteristics and receive returns from economies of specialized production through the standard division of labor and specialization. Consumers then match their preferences to the products available in the market. In the context of golf, course owners attempt to develop a course that suits the characteristics of the golfers in the area and those willing to travel.¹⁰

Section 4 of Rosen's paper provides a test methodology for estimating the theoretical model on both the demand and supply side. As required on the demand side, data are available on characteristics purchased by, in this case golfers, and also their exogenous demand shift variables. On the supply side, data are required on product characteristics, factor prices, and technological differences among sellers. The following represents the demand and supply equations ignoring error terms:

$$p_i(a) = G^i(a_1, \dots, a_n, X_1)(\text{demand}) \quad (1)$$

$$p_i(a) = H^i(a_1, \dots, a_n, X_2)(\text{supply}) \quad (2)$$

where $i = 1, \dots, n$ and where p_i and a_i are all joint dependent price variables and X_1 and X_2 are exogenous demand and supply shifters. Rosen's procedure requires a three-stage process. The first step is to regress observed differentiated product prices on all relevant characteristics without regard to the exogenous shifters. The best fitting functional form should be used in this standard hedonic approach. According to Rosen "This econometrically duplicates the information acquired by agents in the market, on the basis of which they make their decisions" (p. 50).¹¹ Next, we compute a set of implicit marginal prices by differentiating observed product prices by each product characteristic. Simply, observing the coefficients by taking the partial derivative with respect to each characteristic in the standard hedonic equation generates the predicted quality-adjusted prices. Finally, we use the estimated marginal implicit prices as endogenous variables and simultaneously estimate the original demand and supply equations above by replacing the direct prices with the marginal implicit prices. The implicit price results are not immune to the same identification problem that exists when using direct price observations indexes.^{12,13}

Combris, Lecocq, and Visser (1997) estimate a hedonic price equation for Bordeaux wine and attempt to determine if quality matters. Others writing in the hedonic tradition include Do and Grudnitski (1995), who analyze how the selling price of single family residential property is affected when it abuts a golf course. Erikson, Sumka, and Witte (1979), following the empirical model developed by Rosen (1974), estimate a simultaneous equations model for rental housing attributes to account for the heterogeneity of the housing good. Gilley and Pace (1995) address

the problem of the lack of independent variation or multicollinearity seen in the nontraded characteristics of hedonic pricing models. Hartman (1989) evaluates product design and pricing strategies using hedonic methods in the luxury hotel industry. Triplett (1986) discusses how to interpret hedonic results. The survey piece discusses the creation of price indexes for the computer industry and the hedonic work done by the economics department of the IBM Corporation. The high rates of quality change for computers rendered conventional price comparisons obsolete. The use of hedonic methods reflects a more accurate view of quality-adjusted prices. Triplett (1969) also considers the findings by Griliches (1961) that the increases in the automobile component of the consumer price index (CPI) between 1954 and 1960 could be attributed to the substantial quality improvement in automobiles.

Data and Empirical Estimation

Data for this study come from four primary sources: *Sportometrics*,¹⁴ Audubon International, the U.S. Census Bureau, and the Bureau of Labor Statistics. In the *Sportometrics* database, there are over 100 club and course fields describing various characteristics ranging from green fees to dress code policies. The data were collected in 2002 via a survey by *Golf Magazine*. The Audubon International database lists all courses that have completed at least the environmental planning component of certification and all signature certified courses as of 2002. We also obtained publicly available data by county from the U.S. Census Bureau and the Bureau of Labor Statistics on population, income, land area, wet-land area, housing density, employment, and wages.¹⁵

Using the data from Audubon, courses are ranked according to their level of certification as follows:

1. *Environmental planning only*—courses that have completed only the environmental planning aspect of certification,
2. *Environmental planning plus*—courses that have completed environmental planning and at least one of the other components, but not all six categories,
3. *Audubon cooperative sanctuaries*—courses that have completed all six categories, and
4. Audubon signature cooperative sanctuaries.

Summary Statistics

Table 1 reports summary statistics for selected numeric variables in our database. Total golf fees (weekend green fees plus cart fees per 18 holes) are on average greater for certified courses than for noncertified courses. Certified courses are generally newer courses, longer in terms of yardage, and more difficult in terms of slope

Table 1. Summary Statistics

Variable	Number of Observations	Mean	Median	Standard Deviation
All courses				
Total golf fee ^a	9,235	59.60	50.00	32.53
Rounds per day	7,015	110.44	102.80	46.48
Championship yardage	9,235	6,540.86	6,624.00	570.25
Championship slope	9,235	125.46	126.00	9.00
Championship rating	9,235	71.04	71.70	4.56
Course age	9,235	35.81	32.00	26.61
Noncertified courses				
Total golf fee	8,343	57.39	49.00	30.80
Rounds per day	6,328	110.56	102.80	46.47
Championship yardage	8,343	6,524.45	6,608.00	576.60
Championship slope	8,343	124.98	126.00	8.96
Championship rating	8,343	70.93	71.60	4.62
Course age	8,343	36.01	32.00	26.34
Environmental planning only certified courses^b				
Total golf fee	324	72.79	63.00	37.37
Rounds per day	253	112.38	107.98	49.79
Championship yardage	324	6,612.08	6,730.50	637.91
Championship slope	324	129.09	129.00	8.36
Championship rating	324	71.39	72.30	5.79
Course age	324	35.14	29.00	28.84
Environmental planning plus certified courses^c				
Total golf fee	159	73.29	68.00	30.03
Rounds per day	117	111.39	102.24	45.99
Championship yardage	159	6,718.87	6,762.00	293.90
Championship slope	159	128.66	129.00	6.45
Championship rating	159	72.30	72.60	1.57
Course age	159	34.66	29.00	28.99
Audubon sanctuary courses^d				
Total golf fee	385	88.88	85.00	44.35
Rounds per day	301	106.99	102.24	44.64
Championship yardage	385	6,735.20	6,781.00	376.78
Championship slope	385	130.63	131.00	8.25
Championship rating	385	72.44	72.80	2.01
Course age	385	34.40	26.00	28.98
Audubon sanctuary signature courses				
Total golf fee	24	89.17	87.50	35.42
Rounds per day	16	91.06	89.50	27.20
Championship yardage	24	6,988.92	7,023.50	250.03
Championship slope	24	137.79	137.00	5.56
Championship rating	24	73.73	73.80	1.40
Course age	24	4.75	5.00	1.82

a. Calculated as greens fee plus cart fee per 18 holes.

b. Courses that have completed only the environmental planning component of certification.

c. Courses that have completed environmental planning and at least one other category of certification, but not all six.

d. Fully certified Audubon Cooperative Sanctuaries that have completed all six categories.

and course rating. Signature courses have, on average, less rounds played per day on them than on courses with other levels of certification and non-certified courses.

Our empirical analysis includes all public, resort, semiprivate, and private courses that have at least 18 total holes at the facility. The green fee data are the highest seasonal weekend prices for 18 holes.¹⁶ Owing to the fact that green fees at private courses may not necessarily be reflective of market prices, we also estimate the empirical models using only data for public courses.¹⁷

Table 2 shows the total number of courses and certified courses by state. In the United States, there are 1,179 (6.92% of total) courses certified by Audubon at various levels. Of these, 484 (2.84% of total courses) are fully certified sanctuaries. There are 46 signature courses in the United States with 21 of those located in either Florida or California. Delaware, Colorado, New Jersey, Illinois, Florida, Oregon, Maryland, and Connecticut all have at least 10% of their courses certified at various levels. Delaware and Oregon have the largest percentage of fully certified sanctuaries. There are no Audubon courses in Arkansas, the District of Columbia, New Mexico, or Utah.

Hedonic Estimates

Table 3 shows the results of the first step in the Rosen process where we regress the full price of golf (the weekend rate, including cart for 18 holes) on any attribute that might reasonably be linked with quality. We used ordinary least squares (OLS) to regress fees on course yardage, slope, rating, the four levels of certification, and numerous dummy variable course features. The models were estimated in both levels and natural logs.

The single equation certification premiums (in levels) are reported in Table 3.¹⁸ The coefficients for fully certified Audubon Sanctuaries is US\$11.58 for all courses, US\$5.51 for public courses, and highly statistically significant. The price premiums for environmental planning only and environmental planning plus courses are also significant (except for environmental planning only for public courses).¹⁹ These coefficients, however, are not the relevant demand prices, and therein lies the wisdom of the full Rosen implementation. Armed with these coefficients, we now produce quality-adjusted prices to estimate the structural demand and supply equations.²⁰

Supply and Demand Estimates for Golf

The problem with the single equation hedonic price coefficients, as noted by Rosen, is that they do not have any unique or obvious economic interpretation, and that is why he proposed the compelling next step of estimating the structural demand and supply curves. Hence, we now use the predicted values from the single equation first step to simultaneously estimate the structural equations which have reasonable economic interpretation. The model is given in Equations 1 and 2 above. Rounds played per day serves as the measure of quantity and its coefficient should therefore be

Table 2. Audubon International Certified Courses by State

State	Total Courses	Environmental Planning Only	Environmental Planning Plus	Fully Certified Sanctuaries	Signature Courses	Total Certified	Percentage Certified	Percentage Sanctuaries
(1)	(2)	(3)	(4)	(5)	(6)	(3) + ... + (6)	(7)/(2)	(9)/(5)
DE	45	5	2	5	0	12	26.67%	11.11%
CO	253	21	6	24	2	53	20.95%	9.49%
NJ	312	26	5	26	0	57	18.27%	8.33%
IL	738	37	19	43	2	101	13.69%	5.83%
FL	1,319	49	30	78	15	172	13.04%	5.91%
OR	207	4	0	22	0	26	12.56%	10.63%
MD	202	8	8	7	0	23	11.39%	3.47%
CT	182	8	3	8	0	19	10.44%	4.40%
MN	511	22	5	21	3	51	9.98%	4.11%
VA	362	12	3	17	0	32	8.84%	4.70%
NC	595	30	5	16	1	52	8.74%	2.69%
WY	48	1	0	3	0	4	8.33%	6.25%
HI	91	4	0	3	0	7	7.69%	3.30%
CA	1,068	30	13	28	6	77	7.21%	2.62%
NV	116	1	3	2	2	8	6.90%	1.72%
TN	294	10	7	3	0	20	6.80%	1.02%
SC	415	12	5	10	1	28	6.75%	2.41%
NY	887	17	11	23	4	55	6.20%	2.59%
VT	67	2	1	1	0	4	5.97%	1.49%
KS	235	9	2	2	1	14	5.96%	0.85%
PA	752	18	10	16	0	44	5.85%	2.13%
WV	120	1	0	6	0	7	5.83%	5.00%
WA	299	6	2	9	0	17	5.69%	3.01%
IN	485	7	9	11	0	27	5.57%	2.27%
RI	54	2	1	0	0	3	5.56%	0.00%
GA	448	8	2	12	2	24	5.36%	2.68%
MA	356	9	4	6	0	19	5.34%	1.69%
ID	97	2	1	2	0	5	5.15%	2.06%

(continued)

Table 2 (continued)

State	Total Courses	Environmental Planning Only	Environmental Planning Plus	Fully Certified Sanctuaries	Signature Courses	Total Certified	Percentage Certified	Percentage Sanctuaries
MO	354	4	5	7	1	17	4.80%	1.98%
TX	905	16	5	17	1	39	4.31%	1.88%
ND	94	2	0	2	0	4	4.26%	2.13%
MI	959	16	6	17	1	40	4.17%	1.77%
AZ	339	0	6	7	1	14	4.13%	2.06%
NH	101	3	0	1	0	4	3.96%	0.99%
WI	522	11	3	6	0	20	3.83%	1.15%
OH	806	15	6	6	2	29	3.60%	0.74%
KY	287	4	3	3	0	10	3.48%	1.05%
MT	90	2	1	0	0	3	3.33%	0.00%
IA	386	3	0	8	0	11	2.85%	2.07%
ME	119	1	1	1	0	3	2.52%	0.84%
OK	199	2	2	1	0	5	2.51%	0.50%
MS	166	2	1	0	1	4	2.41%	0.00%
SD	89	0	1	1	0	2	2.25%	1.12%
AR	186	2	0	2	0	4	2.15%	1.08%
LA	158	2	1	0	0	3	1.90%	0.00%
AL	288	2	2	0	0	4	1.39%	0.00%
NE	192	1	0	1	0	2	1.04%	0.52%
AK	21	0	0	0	0	0	0.00%	0.00%
DC	6	0	0	0	0	0	0.00%	0.00%
NM	97	0	0	0	0	0	0.00%	0.00%
UT	112	0	0	0	0	0	0.00%	0.00%
Totals	17,034	449	200	484	46	1,179	6.92%	2.84%

Note: AK = Alaska; AL = Alabama; AR = Arkansas; AZ = Arizona; CA = California; CO = Colorado; CT = Connecticut; DC = District of Columbia; DE = Delaware; FL = Florida; GA = Georgia; HI = Hawaii; IA = Iowa; ID = Idaho; IL = Illinois; IN = Indiana; KS = Kansas; KY = Kentucky; LA = Louisiana; MA = Massachusetts; MD = Maryland; ME = Maine; MI = Michigan; MN = Minnesota; MO = Missouri; MS = Mississippi; MT = Montana; NC = North Carolina; ND = North Dakota; NE = Nebraska; NH = New Hampshire; NJ = New Jersey; NM = New Mexico; NY = New York; OH = Ohio; OK = Oklahoma; OR = Oregon; PA = Pennsylvania; RI = Rhode Island; SC = South Carolina; SD = South Dakota; TN = Tennessee; TX = Texas; UT = Utah; VA = Virginia; VT = Vermont; WA = Washington; WI = Wisconsin; WV = West Virginia; WY = Wyoming.

Table 3. Hedonic Equation (Dependent Variable: 18 Hole Weekend Green and Cart Fee)

Variable	All Courses	Public Courses
Intercept	-259.74257*** (99.7558)	-168.39685 (135.5842)
Championship yardage	0.05700*** (0.0222)	0.01219 (0.0212)
Championship yardage squared	-0.00001*** (0.0000)	-0.0000033 (0.000004)
Championship yardage cubed	0.000000001*** (0.0000)	2.62E-10 (0.0000)
Championship slope	8.17271*** (2.5089)	6.12674* (3.4133)
Championship slope squared	-0.09074*** (0.0214)	-0.06757** (0.0288)
Championship slope cubed	0.00033*** (0.0001)	0.00024*** (0.0001)
Championship rating	-1.00532** (0.4260)	0.12177 (1.1379)
Championship rating squared	0.00868*** (0.0032)	-0.00034 (0.0103)
Certification levels		
Environmental planning only	4.5034*** (1.3380)	0.9406 (1.6004)
Environmental planning plus	3.9889* (1.8766)	9.1465*** (2.4658)
Certified sanctuary	11.5799*** (1.2438)	5.5140*** (1.7457)
Signature course	1.5938 (4.8315)	8.1017* (4.9068)
Course Type		
Private	2.1613** (0.8613)	
Semiprivate	-1.5269* (0.7384)	
Resort	25.0400*** (1.0733)	
Selected course controls		
Full bar	6.44236*** (0.9724)	7.37619*** (0.9299)
Pro on site	2.67344 (2.0137)	3.26063* (1.8550)
Caddies available	15.94854*** (0.8947)	12.11779*** (0.1996)
Number of other golf course controls	72	70
Adjusted R ²	0.49	0.46
Observations	9,235	3,547

Note: Standard errors are reported in parenthesis.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

negative in the demand equation and positive in the supply equation. The four dummy variables for the various levels of certification are included as both supply and demand shifters. Various course characteristics are included in the supply and demand equations according to whether we expect them to influence consumer value and/or firm cost. The full bar variable, for example, is included in both the supply and demand equations. The addition of a bar is expected to not only increase demand but also increase costs because there may be cross subsidies, and more importantly, these require management and oversight. Course yardage and course slope are correlated, but we expect slope to be more directly related to maintenance costs. Yardage is therefore included in the demand equation and course slope in supply. Demographic data at the county level such as income (in the demand equation) and worker wages (in the supply equation) are also included in the structural estimates.

Table 4 reports the simultaneous three-stage least square (3SLS) estimates of the supply and demand functions in levels. The coefficient for rounds played per day is negative in the demand equation and positive in the supply equation, supporting the laws of both demand and supply. The coefficients for the following numeric variables are statistically significant and have a positive influence on the demand for golf: hotel employment per state (as a proxy for tourism), county per capita personal income, the average golf fee per county (prices charged by other courses in proximity—a measure of the price of substitute courses), proportion of county considered wet-land, and county population per square mile. Golf course characteristics that have an estimated positive influence on demand are Audubon certification (not including signature courses), resort courses, courses that offer full bars, and courses that have caddies available.

On the supply side, the characteristics that positively influence costs are courses that provide full bars, have golf professionals on-site, offer apparel and golf clubs in the pro shop, have caddies available, have homes on courses, aerate more than three times per year, and are considered resort courses. In addition, there is a decrease in supply for courses that are certified in environmental planning only, environmental planning plus, and fully certified courses. The average rainfall per state and the proportion of the county considered wet-land both have negative and significant coefficients in the supply equation (that is, positive shifters of supply), indicating that these two factors may reduce irrigation costs. One counterintuitive negative (but statistically insignificant) supply coefficient is the county average wage per job, included as a proxy for labor costs.²¹ In the natural logs estimation, however, this coefficient is positive and significant at the 1% level, providing an intuitive decrease in supply due to increased labor costs.

The 3SLS supply and demand estimates in natural logs are reported in Table 5.²² The coefficients for rounds per day in the demand equation indicate an elasticity of demand of -1.19 for all courses and -1.65 for public courses.²³ A more elastic demand for public courses is not surprising considering that golfers at public courses are probably more price sensitive and also since there are more substitutes available for any one particular public course. The elasticity of supply is 19 for all courses and 9 for public courses. A relatively flat supply curve for golf indicates that there are

Table 4. Supply and Demand Functions (Dependent Variable: Predicted Total Golf Fees From Levels Hedonic)

Variable	All Courses	Public Courses
	Demand	
Intercept	99.7739*** (10.7807)	43.9404*** (11.1073)
Rounds per day	-0.4842*** (0.0319)	-0.2945*** (0.0260)
Champ yardage	-0.0321*** (0.0041)	-0.0116*** (0.0043)
Champ yardage squared	0.000004*** (0.0000)	0.000002*** (0.0000)
State hotel employment	0.0007*** (0.0001)	0.0004*** (0.0001)
Per capita personal income	0.0004*** (0.0000)	0.0003*** (0.0001)
State average rainfall	-0.0040** (0.0016)	-0.0038* (0.0020)
County average golf fee	0.2968*** (0.0206)	0.3525*** (0.0257)
Environmental planning only	6.6267*** (1.6158)	4.4553*** (1.8985)
Environmental planning plus	7.4093*** (2.3403)	12.4284*** (3.2889)
Certified sanctuary	14.7232*** (1.4996)	6.8400*** (1.9891)
Signature course	0.5623 (6.2968)	6.3272 (6.0802)
Full bar	9.4085*** (0.7427)	9.9681*** (0.7120)
Pro on site	5.8201*** (1.2520)	5.5616*** (1.1231)
Caddies available	14.3793*** (1.2374)	17.9033*** (2.2479)
Homes on course	-1.0503 (0.7180)	-1.1855 (0.7540)
Resort course	24.6603*** (1.1887)	
Pro shop products	2.3224 (1.4516)	1.4764 (1.2250)
No overseeing	-0.5817 (0.6309)	-1.1904* (0.7185)
Proportion county wet-land	6.0360*** (2.0820)	5.3784*** (2.3158)

(continued)

Table 4 (continued)

Variable	All Courses	Public Courses
Population per square mile	0.0011*** (0.0002)	0.0006*** (0.0002)
Supply		
Intercept	373.4043*** (14.5039)	390.8847*** (23.2899)
Rounds per day	0.1691*** (0.0224)	0.1087*** (0.0259)
Course age	-0.0703*** (0.0226)	-0.2691*** (0.0416)
Course age squared	0.0012*** (0.0002)	0.0023*** (0.0004)
Championship slope	-7.2804*** (0.2495)	-6.9798*** (0.3961)
Championship slope squared	0.0355*** (0.0010)	0.0330*** (0.0016)
County average wage per job	-0.000005 (0.0000)	-0.00004 (0.0001)
Number of courses in county	0.0322*** (0.0046)	0.0362*** (0.0077)
State average rainfall	-0.0019*** (0.0006)	-0.0033*** (0.0009)
Environmental planning only	4.2895*** (0.8412)	-0.8591 (1.2188)
Environmental planning plus	6.0843*** (1.2109)	7.9467*** (2.0539)
Certified sanctuary	14.6028*** (0.7734)	8.5764*** (1.2278)
Signature course	3.1882 (3.2553)	5.4773 (3.7699)
Full bar	10.6412*** (0.3887)	8.2276*** (0.4443)
Pro on site	3.2364*** (0.6566)	1.2141 (0.7380)
Caddies available	20.9633*** (0.6681)	12.6345*** (1.4123)
Homes on course	4.6290*** (0.3844)	0.1631 (0.4634)
Aerate three times per year or more	4.1399*** (1.1396)	3.9717* (2.2976)
Resort course	30.3242*** (0.5810)	
Pro shop products	2.3030*** (0.7495)	-0.3446 (0.7674)

(continued)

Table 4 (continued)

Variable	All Courses	Public Courses
No overseeing	-0.3845 (0.3272)	-0.8405* (0.4494)
Proportion county wet-land	-8.3153*** (1.1488)	-2.6025* (1.5071)
Observations	7,015	2,509
System-weighted R^2	0.84	0.73

Note: Standard errors are reported in parenthesis.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

few specialized inputs in the golf industry. The result is fairly intuitive considering that specialized land, labor, and capital are typically not needed in the construction of most golf courses.

Examining the certification coefficients in both the supply and demand estimates indicates that certification increases the demand and decreases the supply for golf. In the 3SLS levels estimates, the upward shift in demand for fully certified sanctuaries is US\$14.72 (for all courses) and is slightly greater than the downward shift in supply estimate of US\$14.60. In the logs estimates, supply decreases by 21% for certified sanctuaries and demand increases by 18%. For the less demanding levels of certification, demand increases by slightly more than the decrease in supply in both the levels and log estimations.²⁴ Signature courses have no statistical impact on supply and demand in the levels model. In the natural logs estimation, there is a significant downward shift in supply but no significant upward shift in demand for signature courses. This result is consistent with the notion that some new courses may be pressured into or even required to obtain signature status before opening a new course. Signature courses do have a positive influence on demand when estimating the model for public courses only.

Taking the single equation hedonic estimate of certification on price and comparing it to the structural estimates reveals two different pictures and suggests why Rosen's proposed approach is valuable. First, the coefficient estimates on willingness to pay are different; the structural estimate on demand is about US\$3.00 or 25% higher than the single equation method. Second, according to the estimates from the supply equation, there are costs associated with producing this extra quality.

Conclusion

We have assembled a large database on golf courses in the United States, which measures many physical and economic features. With these data, we have engaged

Table 5. Supply and Demand Functions in Natural Logs (Dependent Variable: Predicted Total Golf Fees From Logs Hedonic)

Variable	All Courses	Public Courses
	Demand	
Intercept	-4.4081*** (0.4214)	-3.2224*** (0.5173)
Rounds per day ^a	-0.8392*** (0.0435)	-0.6077*** (0.0455)
Champ yardage ^a	0.9837*** (0.0445)	0.7761*** (0.0525)
State hotel employment ^a	0.0295*** (0.0096)	-0.0054 (0.0125)
Per capita personal income ^a	0.1955*** (0.0227)	0.1865*** (0.0312)
State average rainfall ^a	-0.00003 (0.00003)	-0.00001 (0.00003)
County average golf fee ^a	0.2630*** (0.0187)	0.2694*** (0.0252)
Environmental planning only	0.1085*** (0.0232)	0.1141*** (0.0331)
Environmental planning plus	0.1548*** (0.0337)	0.2453*** (0.0577)
Certified sanctuary	0.1826*** (0.0217)	0.1094*** (0.0348)
Signature course	0.1126 (0.0906)	0.2417** (0.1063)
Full bar	0.2107*** (0.0107)	0.2130*** (0.0125)
Pro on site	0.1667*** (0.0180)	0.1374*** (0.0194)
Caddies available	0.1675*** (0.0174)	0.2160*** (0.0393)
Homes on course	-0.0063 (0.0100)	-0.0143 (0.0131)
Resort course	0.3070*** (0.0171)	
Pro shop products	0.0705*** (0.0209)	0.0374 (0.0215)
No overseeing	0.0112 (0.0091)	0.0029 (0.0126)
Proportion county wet-land ^a	0.0128*** (0.0030)	0.0138*** (0.0041)
Population per square mile ^a	0.00002*** (0.0000)	0.000005 (0.00000)

(continued)

Table 5 (continued)

Variable	All Courses	Public Courses
	Supply	
Intercept	-7.7894*** (0.2106)	-4.6205*** (0.2317)
Rounds per day ^a	0.0527** (0.0234)	0.1080* (0.0352)
Course age ^a	0.0054* (0.0023)	-0.0598*** (0.0055)
Championship Slope ^a	2.2770*** (0.0303)	1.6441*** (0.0396)
County avg. wage per job ^a	0.000002*** (0.00000)	0.000002*** (0.00000)
Number of courses in county ^a	0.0186*** (0.0023)	0.0128*** (0.0038)
State average rainfall ^a	-0.00002*** (0.0000)	-0.0001*** (0.00001)
Environmental planning only	0.0845*** (0.0094)	0.0368** (0.0160)
Environmental planning plus	0.1297*** (0.0136)	0.1376*** (0.0275)
Certified sanctuary	0.2110*** (0.0087)	0.1455*** (0.0164)
Signature course	0.1720*** (0.0367)	0.2047*** (0.0505)
Full bar	0.2184*** (0.0044)	0.1893*** (0.0059)
Pro on site	0.1137*** (0.0074)	0.0561*** (0.0098)
Caddies available	0.2824*** (0.0078)	0.1730*** (0.0187)
Homes on course	0.0525*** (0.0040)	-0.0004 (0.0061)
Aerate 3 times per year or more	0.0642*** (0.0143)	0.1355*** (0.0304)
Resort course	0.3815*** (0.0065)	
Pro shop products	0.0492*** (0.0085)	0.0020 (0.0103)
No overseeing	0.0145*** (0.0037)	0.0031 (0.0060)
Proportion county wet-land ^a	-0.0068*** (0.0012)	0.0021 (0.0019)
Observations	7,015	2,509
System-weighted R ²	0.81	0.67

Note: Standard errors are reported in parenthesis.

a. Natural log.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

the method proposed by Rosen in 1974 to measure quality-adjusted structural demand and supply equations for golf. Although a simple, one equation price estimation of quality parameters may be appropriate in some cases, the interpretation of the coefficients can be difficult or misleading. The two-step Rosen approach overcomes this hazard and allows for straightforward interpretation of the structural equations.

Although the premiums for certification are quite large and it may seem implausible that golfers themselves are willing to pay substantially more to play an environmentally friendly course, we argue that the results of this analysis show that there is a real demand for environmental quality on golf courses. This demand may come in the form of playing a course with an increased natural environment or the simple personal satisfaction of knowing that the course is environmentally friendly. Likewise, wearing a badge of certification may be a signaling device used to convey quality, certain ideas, or attitudes. In the end, whatever forces are driving the demand, certification appears to be a simple means of communicating the stewardship activities of a golf course, whether it is to golfers or third parties. We find that golfers pay about 10% to 18% more per round to play environmentally certified golf courses.

Notes

1. Others have claimed environmental benefits for golf courses (Nelson, 1997) including reduced noise pollution, green open space, water purification, temperature modification, and increased wildlife habitat. See also Limehouse and McCormick (2007) for more on golf courses as open green space.
2. A 1982 Environmental Protection Agency (EPA) report concluded that, on average, a golf course uses nine pounds of herbicides, fungicides, and insecticides per acre per year—three times the amount used by the most chemically intensive agribusiness (Selcraig, 1993).
3. There is of course the well-established literature on brand names tracing to Klein, Crawford, and Alchian (1978). In our view, certification acts in the same manner.
4. We are working on this issue as a separate topic. See Limehouse and McCormick (2007).
5. During construction of the Clemson University Walker Course in the 1990s, several environmental groups opposed construction of the course citing chemical runoff into Lake Hartwell adjoining the course and decreased wildlife habitat. Several individuals at Clemson University took an interest in the Audubon program to show that the course could benefit the environment with proper management and assistance from Audubon International (D. Garrett, personal communication, 2003).
6. Audubon International is not affiliated with the National Audubon Society. Although golf is the focus of this analysis, it should be noted that Audubon International also certifies businesses, parks, schools, and cemeteries. Audubon International is a not-for-profit organization specializing in conservation outreach assistance, education, and research.
7. We have performed no formal research on the costs of certification. However, Audubon has conducted several case studies with participating courses. Carolina National Country

Club in Bolivia, North Carolina, naturalized 30 acres to create wildlife habitat as part of its participation in the Audubon Sanctuary program. Areas naturalized were either maintained turf or areas set aside during the construction phase of the course for naturalization. The club estimated that it cost US\$10,000 to implement the project and that it saves US\$15,000 annually. Savings come in the form of decreased man-hours, fertilizer, and wear and tear on equipment needed to maintain these areas. Carolina National Country Club points out that one important aspect of the program is communicating their participation in the program to golfers with signs throughout the course and brochures in the pro shop. This suggests that courses find it important to communicate its environmental stewardship efforts to golfers who may demand playing golf on an environmentally friendly certified course.

8. As computational costs have decreased, economists have increasingly used hedonic methods to analyze the pricing of both durable and nondurable goods. The automotive industry is by far the most widely analyzed. Cowling and Cubbin (1972) worked on the United Kingdom motor industry, De Pelsmacker (1988) on the Belgian car market, and further studies of the United States automotive industry have been done by Griliches (1961), Hogarty (1975), and Triplett (1969), among others. More recently, hedonic pricing models have been used to analyze the computer industry, housing, clean air, and real estate.
9. In work similar to ours in spirit and substance, but with a smaller database, Krock (1999) models the supply and demand of public golf in Chicago and uses techniques from the urban economics, hedonic pricing, and spatial competition fields.
10. Rosen assumes there are a sufficiently large number of differentiated products in the market so the available product characteristic combinations are continuous for all practical purposes. He points out that this assumption was originally made by Court (1941) and allows the use of marginal analysis rather than programming methods as required by Lancaster (1966). Here, given the amount of data available, a continuous spectrum of course characteristics is not unreasonable. The only possible drawback is that not all regions or markets will have golf courses that suit area golfers' consumption preferences. Another related assumption is that there are no secondhand markets, or alternatively, that goods represent pure consumption. Both cases apply to golfers. Deviating from Lancaster (1966), our model assumes products are not divisible. Indivisibility means that a golfer cannot purchase the right to play on one course's fairways and putt on another course's greens while eating in another's restaurant. When a golfer pays her green fee, she is purchasing a bundled, pure consumption good. Another theoretical finding in the hedonic model is that there is a natural tendency toward market segmentation. This applies where golfers can decide on whether to become a member of a private club with its corresponding privileges or play at daily fee clubs. Private club privileges include less course congestion, a venue to entertain clients, networking opportunities, and reputation effects. A private membership also signals the individual is committed or bonded to the area.
11. In this step, the product price is effectively regressed on any and all variables that might reasonably be expected to affect price. The primary focus of this step is to create quality-adjusted, predictions of price based on the observed data.
12. Ekeland, Heckman, and Nesheim (2004) address criticisms of Rosen's two-stage technique and much of the ambiguity in the literature concerning whether technology and

consumer preferences can be properly identified in single hedonic markets. Epplé (1987), for example, questions the parameter estimates when quality-adjusted prices derived from hedonic models are used as endogenous variables to simultaneously estimate demand and supply using OLS. Although the criticisms of Rosen's technique are not a major focus of the analysis presented here, it should be noted that Ekeland et al. (2004) find that many of these criticisms are unwarranted.

13. Rosen goes on to discuss four possible identification problems. The first occurs when all buyers are different but cost conditions are identical across firms. This would cause the supply shift variables to drop out of the supply equation and therefore the sample observations on implicit prices and characteristics identify compensated supply functions. The second happens when buyers do not differ but sellers do. This is a mirror image to the situation above so demand shifters would fall out of the demand equation and the sample estimates compensated demand functions. The third possibility occurs when both buyers and sellers are identical. In this situation, the consumption and production choices, represented by offer curves in the Rosen model, are tangent at a single point and no product differentiation is present in the market. The final and most realistic possibility is that both buyers and sellers fall within their own distinct distribution. In this case, both demand and supply shifters have nonzero variance.
14. *Sportometrics* is a golf course consulting company who licensed their data to us for this project.
15. Population, income, and density data are available at the U.S. Census Bureau's American Fact Finder website: <http://factfinder.census.gov/>. Employment and wage data are available at the Bureau of Labor Statistics website: <http://www.bls.gov/>.
16. A large number of courses include their cart fee with green fee and do not list a separate cart fee. This provides a combined green and cart fee. For other courses where prices are unbundled, we added cart fees to green fees to create an equivalent variable (total golf fees) to use in the analysis. We exclude courses that do not offer carts since their reported green fees are not comparable. We note, however, that we have estimated our models with these courses and get almost identical results. We also ran our models using the weekday (as opposed to weekend) quoted prices. These results are almost identical as well.
17. See Shmanske (1998) for details on various pricing practices for different segments of the golf market.
18. The coefficients for the individual dummy variable course characteristics are not reported for brevity. These results, however, show that golfers pay more to play a round of golf at courses with yardage markers, Poa Annua Grass greens, caddies available, with strict dress codes, and courses with certain training facilities.
19. One surprising finding is that there is no price premium for signature courses (presumably the highest and most stringent level of certification). One explanation may be that allowing environmentalists to play an active role in the planning of a golf course alters the sport characteristics of the course in ways that reduce the players' enjoyment of the golf course.
20. We also estimated the single equation hedonic price model natural logs. The coefficients for the certification variables indicate that sanctuary courses charge a 15.7% premium

and that environmental planning only and plus courses charge premiums of 7.00% and 8.60%, respectively. The coefficient on signature courses becomes significant in the logs equation and indicates a premium of 13.5%. The significance of other course characteristics is almost identical to that of the levels hedonic equation.

21. In an alternative model, we include the state average wage for landscaping workers instead of the county average wage per job. This model yields the same counterintuitive result and has a very little impact on the remaining coefficients.
22. The first stage here (for the supply and demand estimates in Table 5) is the logs equation. That is, we are using the predicted values from the log hedonic equation.
23. Because price (as opposed to quantity) is the dependent variable, the elasticity from the log equation is calculated as the reciprocal of the coefficient on rounds per day.
24. We note that lower levels of certification do not appear to cost much and yet are both statistically and financially significant. In fact, for public courses, the premiums for environmental planning plus are higher than for full certification. This suggests that this new phenomena in the golf industry has not been fully absorbed by investors in this market. We expect that this result follows from the recent introduction of this product and that soon additional courses certified at lower levels will drive down the apparent price premium.

Appendix

Audubon Certification Levels and Requirements

Environmental Planning—This is the initial step required for certification. Once the plan is accepted by Audubon, the golf course receives certification in environmental planning and is considered a registered member. The plan is designed to assist courses with outlining projects and management strategies in various areas of the course including greens, tees, fairways, rough, equipment, irrigation, landscaping, and natural areas. The plan must be centered on improving environmental practices and implementing environmental programs in each of the five remaining program areas. The environmental plan itself does not require environmental action by the golf course. Instead, the course manager or superintendent must complete a five section (one for each of the remaining areas of the program) self-assessment of environmental management practices. The course must state its planned efforts for each management practice required by Audubon International that is not already fully implemented.

Wildlife and Habitat Management—This category of certification focuses on enhancing natural areas and landscaping the course so as to improve the habitats of native wildlife. Aspects of this category include understanding and identifying the needs of wildlife, landscaping with native plants, protecting woodlands and wetlands, naturalizing nonplay turf areas, and mounting nest boxes. Photographs of naturalized areas, nest boxes, habitat signs, and the like, are required. A wildlife inventory of bird and mammal species on the course must also be produced. The

course must also submit a course map showing the layout of the golf holes, water bodies, naturalized areas, no spray zones, and wildlife structures.

Chemical Use Reduction and Safety—In addition to complying with all Occupational Safety and Health Administration (OSHA) regulations, member courses are expected to implement Best Management Practices and an extensive Integrated Pest Management Program. Best Management Practices involve exercising precautions before using chemicals and minimizing the amount used. Integrated Pest Management Programs are management procedures used to reduce or eliminate turfgrass pest populations without pesticide application. Integrated Pest Management Programs include proper plant selection, mowing heights, fertility management, soil testing, and proper drainage. This area of certification also places a strong emphasis on the maintenance facility and requires careful storage of chemicals, proper location of mix and fuel storage, and extensive training of employees. Audubon International requires photographs of the chemical storage area, chemical mix and load area, equipment storage area, and equipment wash area. The course must also submit reference charts that detail aesthetic, cultural, fertilizer, and pesticide practices.

Water Conservation—This area focuses on ensuring adequate water supplies for irrigation and healthy ecological functions surrounding golf courses. The key components of this plan are the installation and maintenance of efficient irrigation systems, use of ground water, weather data, the elimination of all unnecessary water use, reduction of the number of acres irrigated, and minimization of water loss from runoff and evaporation. Courses are also encouraged to select varieties of turfgrass and plants that are drought tolerant or require less irrigation.

Water Quality Management—This category is closely related to the chemical use reduction category in that both emphasize reducing chemical runoff into the water supply. Water quality management, however, focuses on the health of oceans, rivers, streams, wetlands, lakes, and ponds around or on golf courses. Specific components include erosion control, vegetated buffers, and retention ponds. Best Management Practices and/or structural controls around bodies of water must be near all bodies of water to reduce chemical runoff. Courses must submit photographs of water bodies and results of chemical water tests. In addition, courses are required to submit reference charts that document management practices for each body of water on the course.

Outreach and Education—The outreach and education component of the program ensures community involvement and increases public awareness of environmental stewardship. Golf courses are encouraged to host community nature walks, visit local schools to help students develop wildlife gardens on school properties, and post detailed signs around naturalized areas describing the project. Each course must form a Resource Advisory Group to help plan environmental projects and educational programs. Photographs of Audubon Sanctuary displays and people involved in outreach activities are required. Member courses must also document their educational and outreach activities by submitting copies of news articles, brochures, posters, and so on.

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